
PolyTwist

A Stratified Kinematic Atlas for Cut-and-Turn Mechanisms

Exact geometry, partial dynamics, and proof-carrying benchmark semantics

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Canonical repository audit base: a970c65ab367b98ea380d48919c449bb63fa8638.

This monograph changes research mathematics and architecture documentation only. It does not claim that the current software implements the atlas developed here.

Research status and provenance

Claim classes used throughout

Tag	Meaning
Established	Previously established mathematics, cited to primary or authoritative literature.
Definition	A repository-specific object. A definition is not a novelty claim.
Proved here	A theorem proved in this monograph from explicitly stated assumptions.
Evidence	A deterministic computation or exact instance check that has not been promoted to a general theorem.
Conjecture/open	A statement or algorithmic target without a complete proof.

Every substantial statement is mirrored in `research/claims.json`; bibliographic metadata and claim support are mirrored in `research/sources/registry.json`. The monograph deliberately separates three things that technical writing often tosses into one suspiciously convenient bucket: established mathematics, PolyTwist definitions, and actual new proofs.

Boundary of the claim

The phrase *canonical representation* has two levels. The mathematical object is canonical as an isomorphism class of exact atlases. A practical byte-for-byte canonical serializer under all geometric symmetries is not proved here; it is isolated as theorem 8.3.

Abstract

A cut-and-turn puzzle is not a mesh, a sticker permutation, or a list of move names. It is a geometric mechanism whose cuts induce a stratified body, whose bonds induce physical pieces, whose assembly constraints induce a configuration space, and whose legal motions induce partial dynamics. Existing PolyTwist code already enforces a valuable principle – pixels observe mechanics rather than define them – but its canonical state is still a cubic logical lattice with convex geometry attached afterward. That is sufficient for Rubik-like shape modifications and insufficient for arbitrary mechanisms.

This monograph proposes the *Stratified Kinematic Atlas*

$$\mathfrak{P} = (\mathbb{K}; B, \Phi, \beta, \mathcal{J}, \mathcal{T}, O)$$

as the canonical mathematical object. The compact semialgebraic body B and oriented cut functions Φ induce a finite connected sign-component stratification. A bond system β quotients atomic chambers into rigid physical pieces. Exact assembly data $\mathcal{J}$ derive the continuous configuration space and docked state space. A turn template specifies an anchor, a gate, a semialgebraic rigid-motion path, and a docking relation. A move is legal precisely when it satisfies gate atomicity, bond closure, swept-path admissibility, and endpoint docking.

The resulting state transitions are partial bijections, not necessarily total permutations. They generate an inverse semigroup and an associated legal-motion groupoid; the usual Rubik group action appears exactly when every primitive move is total. This yields a single derivation chain from cuts to pieces, legality, state graph, scrambles, planning, rendering, and active system identification. The monograph proves finiteness and specialization results, exact trace-clipping formulas, a bond-incidence criterion, semialgebraicity of turn domains, finite legality stratification, isomorphism invariance, a passive-observation nonidentifiability theorem, and exact product-automaton scramble laws. Two rational ghost-like cut atlases are worked out with exact matrices, chamber witnesses, primitive plane equations, trace endpoints, and TikZ diagrams.

The novelty claim is intentionally narrow. Hyperplane arrangements, configuration-space motion planning, partial actions, permutation puzzle formats, and plane-cut puzzle generators all predate PolyTwist. The plausible research contribution is their exact integration into a proof-carrying mechanism specification and a benchmark whose observations and labels are derived from that same object.

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Chapter 1

Why the puzzle itself needs a mathematical object

1.1 The research target

The goal is not another Rubik’s Cube benchmark with more textures and a heroic amount of JSON. The target is a mathematical language capable of expressing, from one exact object,

- affine, curved, intersecting, offset, and procedurally generated cuts;
- visible traces and their metric and topological relations;
- the induced physical pieces, including bandages and hidden joins;
- continuous mechanical configurations and discrete docked states;
- state-dependent legal and illegal motions;
- the induced state graph, scramble laws, planning problems, and experiments;
- exact surfaces, certified meshes, and rendered observations.

These are not independent modules. They are successive images of one mechanism object. If geometry says one thing and the state engine says another, the benchmark has become a stage prop.

1.2 Repository audit at the canonical base

The canonical repository was audited at commit `a970c65ab367b98ea380d48919c449bb63fa8638` [Li26]. The current system is technically disciplined within its scope:

- mechanics are exact integer-valued transforms on a cubic logical lattice;
- geometry is compiled from convex half-space intersections;
- a displaced and rotated frame produces Ghost-Cube-like shape modifications;
- bandages make the legal action mask state-dependent;
- rendering and benchmark export observe the exact engine state.

The limitation is foundational. The compiler begins with an $N \times N \times N$ lattice, three orthogonal coordinate families in one frame, and quarter-turn generators. Physical pieces are intersections of an outer convex hull with those pre-existing lattice cells. The engine selects a move by an integer coordinate layer and checks only whether a declared rigid bandage is split. Swept geometric collision, arbitrary cut carriers, non-cubic bodies, non-quarter-turn joints, and jumbling are not represented by the canonical state.

The existing mathematical ledger correctly labels its Slice-Action Code and Unified Legality Operator as provisional. They are useful diagnostics, but neither can reconstruct the mechanism that produced them. This monograph preserves the correct affine, support, and scramble results while replacing the foundation.

1.3 Cuts alone are not a puzzle

A tempting definition is “body plus cuts.” It fails before the notation has time to become impressive.

Theorem 1.1 (Cuts underdetermine dynamics [THM-CUTS-UNDERDET-001]). *There exist two cut-and-turn mechanisms with identical body, oriented cut carriers, chamber decomposition, and physical pieces, but nonisomorphic legal-motion systems.*

Proof. Let $B = [-1, 1]^3$ and use the three oriented planes $x = 0$, $y = 0$, and $z = 0$. With no bonds, the atomic chambers are the eight open octants clipped by B .

Mechanism M_0 declares no turn templates. Its state graph contains only isolated states.

Mechanism M_1 declares one turn a : select the four chambers in $x > 0$ and rotate them together by $\pi/2$ about the x -axis. During the path, the selected interiors remain in $x > 0$ and the stationary interiors remain in $x < 0$, so no selected interior meets a stationary one. The right half-cube is invariant under the quarter-turn, hence the endpoint permutes its four octants and is docked. The reverse quarter-turn is legal.

Thus M_1 contains a nontrivial order-four partial transformation while M_0 contains none. Their geometric cut data agree and their dynamics do not. $\square$

Design rule

Kinematics are primitive atlas data constrained by geometry. They cannot be inferred uniquely from cut traces, even with perfect geometric measurement.

1.4 The derivation principle

The desired architecture is summarized in figure 1.1. Every arrow must be deterministic and independently verifiable.

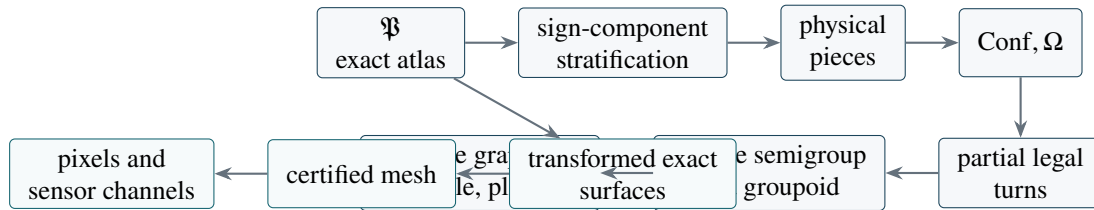


Figure 1.1: The canonical derivation chain. Rendering and benchmark dynamics share the same source object instead of politely disagreeing in separate subsystems.

Chapter 2

Exact substrate and the Stratified Kinematic Atlas

2.1 Effective exact coefficients

The mathematical semantics live over $\mathbb{R}$. An encoded instance uses an *effective ordered coefficient domain* $\mathbb{K} \subset \mathbb{R}$ with exact arithmetic and decidable sign. The default implementation target is the field of real algebraic numbers; rational instances form an especially pleasant subset because humans occasionally deserve nice denominators.

A subset of $\mathbb{R}^n$ is *semialgebraic over* $\mathbb{K}$ when it is a finite Boolean combination of polynomial equalities and strict or weak inequalities with coefficients in $\mathbb{K}$. Semialgebraic sets are closed under Boolean operations and projection, have finitely many connected components, and admit compatible finite triangulations and cell decompositions [BPR06; BCR98]. Quantifier elimination over real closed fields makes these closure statements algorithmic in principle [Col75].

Definition 2.1 (Regular-closed body). A *body* is a compact semialgebraic set $B \subset \mathbb{R}^3$ satisfying

$$B = \text{cl}(\text{int } B), \quad \dim B = 3.$$

Regular closure excludes free-floating lower-dimensional debris from the canonical solid.

2.2 Exact rigid paths

A placement is an element of $\text{SE}(3)$, represented by (R, t) with $R \in \text{SO}(3)$ and $t \in \mathbb{R}^3$. The defining equations $R^T R = I$ and $\det R = 1$ are polynomial, so $\text{SE}(3)$ is semialgebraic.

Ordinary trigonometric parameterizations are not semialgebraic. This is not a reason to abandon exact paths; it is a reason to parameterize them properly.

Lemma 2.2 (Rational half-angle rotation path). *Let $u \in \mathbb{K}^3$ be a unit vector and let U be its cross-product matrix. For a parameter s define*

$$R_u(s) = I + \frac{2s}{1+s^2}U + \frac{2s^2}{1+s^2}U^2.$$

Then $R_u(s) \in \text{SO}(3)$ and represents rotation about u through angle $2 \arctan s$. If an endpoint rotation has algebraic sine and cosine and is not a half-turn, it admits such an exact semialgebraic path. A half-turn is the concatenation of two quarter-turn paths.

Proof. Substitute $\sin \theta = 2s/(1 + s^2)$ and $1 - \cos \theta = 2s^2/(1 + s^2)$ into Rodrigues' formula. The matrix identities $U^T = -U$, $U^2 = uu^T - I$, and $U^3 = -U$ give $R_u(s)^T R_u(s) = I$ and determinant one. For an endpoint with $\cos \theta \neq -1$, take $s = \sin \theta/(1 + \cos \theta)$. Algebraicity is preserved by field operations. Split $\theta = \pi$ into two paths of angle $\pi/2$. $\square$

2.3 The canonical tuple

Definition 2.3 (Stratified Kinematic Atlas [DEF-SKA-001]). A *Stratified Kinematic Atlas* (SKA) is a tuple

$$\mathfrak{P} = (\mathbb{K}; B, \Phi, \beta, \mathcal{J}, \mathcal{T}, O)$$

with the following data.

1. $\mathbb{K}$ is an effective exact coefficient domain.
2. $B \subset \mathbb{R}^3$ is a compact three-dimensional regular-closed semialgebraic body.
3. $\Phi = (\phi_1, \dots, \phi_m)$ is an ordered family of continuous semialgebraic functions on a neighborhood of B . Each ϕ_i is oriented: multiplying it by a positive scalar preserves the cut, while multiplying by a negative scalar reverses its sides.
4. β is an admissible bond system on the atomic chambers induced by (B, Φ) .
5. $\mathcal{J}$ is exact semialgebraic assembly data: fixed bodies, anchors, joints, clearance and nonpenetration relations, global gauge, intrinsic piece symmetries, and a docking predicate.
6. $\mathcal{T}$ is a finite family of reversible turn templates.
7. O is exact appearance and sensor data attached to bodies, surfaces, and strata.

Every cut carrier is required to have empty interior in B ; equivalently, no ϕ_i vanishes on a nonempty open subset of B .

The tuple separates three layers without divorcing them. (B, Φ, β) derives the physical solids. $\mathcal{J}$ derives admissible and docked placements of those solids. $\mathcal{T}$ derives legal state transitions through exact paths. O derives observations and can be changed for appearance-transfer experiments without secretly rewriting mechanics.

2.4 What is canonical

The atlas is not canonical because its cuts have cute IDs. It is canonical as an exact mathematical structure up to rigid isomorphism, defined in theorem 8.1. Coefficient normalization, deterministic ordering, and graph canonicalization are serialization problems layered on top of that structure.

This distinction matters. A JSON digest can be deterministic and still encode the wrong mathematical object. Conversely, two byte-distinct exact files can define isomorphic mechanisms.

Chapter 3

Canonical cut stratification

3.1 Connected sign components

For $x \in B$, define the ternary sign vector

$$\sigma(x) = (\text{sgn } \phi_1(x), \dots, \text{sgn } \phi_m(x)) \in \{-, 0, +\}^m.$$

For a fixed sign vector σ , let

$$X_\sigma = B \cap \bigcap_{i=1}^m \{x : \text{sgn } \phi_i(x) = \sigma_i\}.$$

Definition 3.1 (Connected sign-component stratification [DEF-SIGNCOMP-001]). The canonical cut stratification $\mathcal{X}(B, \Phi)$ is the set of connected components of all nonempty X_σ , each equipped with its sign label and the closure-incidence relation

$$A \preceq C \iff A \subseteq \text{cl } C.$$

The component suffix is essential. A sign vector says which side of every carrier contains a point; it does not say which connected region contains the point.

Theorem 3.2 (Finiteness [THM-SIGN-FINITE-001]). *The stratification $\mathcal{X}(B, \Phi)$ is finite.*

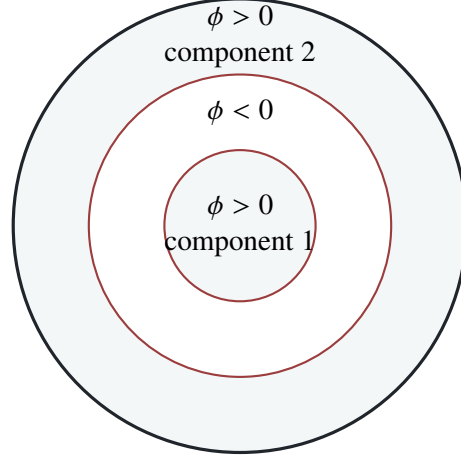
Proof. Every X_σ is semialgebraic because it is a finite intersection of semialgebraic sign conditions with B . A semialgebraic set has finitely many connected components. There are only 3^m sign vectors. $\square$

3.2 Atomic chambers and lower-dimensional strata

The *atomic chambers* are the connected components of

$$B^\circ \setminus \bigcup_{i=1}^m \phi_i^{-1}(0).$$

They are exactly the three-dimensional strata with strict sign labels. Lower-dimensional strata encode carrier intersections, boundary traces, vertices, and degeneracies. The closure-incidence poset records how all of these pieces meet.



Cross-section with $B = \{r \leq 3\}$ and $\phi = (r^2 - 1)(r^2 - 4)$. The positive sign condition has two connected components. A bare sign vector merges them and loses topology.

Figure 3.1: Why connected components, not sign vectors alone, are canonical.

Proposition 3.3 (Chamber cover). *Under the empty-interior assumption on the cut carriers, the closures of the atomic chambers cover B , and distinct chambers have disjoint interiors.*

Proof. The finite union of carrier zero sets is closed and nowhere dense in B . Its complement is dense in B and is the disjoint union of the atomic chambers. Taking closures yields B . Distinct connected components of an open set are disjoint, so their interiors are disjoint. $\square$

3.3 The affine-convex specialization

Let B be convex and let every cut be affine,

$$\phi_i(x) = n_i^T x - d_i.$$

Theorem 3.4 (Affine-convex sign completeness [THM-AFFINE-CONVEX-001]). *Every nonempty strict sign region X_σ is convex and therefore connected. Hence strict sign vectors index atomic chambers without an additional connected-component identifier.*

Proof. A strict affine sign condition is an open half-space. Its intersection with the convex set B and finitely many other half-spaces is convex. Every nonempty convex set is connected. $\square$

This theorem identifies the exact boundary of the current repository's sign-vector intuition. It is correct for convex bodies and affine carriers. It is not a universal law of puzzle geometry.

3.4 The oriented-matroid shadow

For an affine arrangement in all of $\mathbb{R}^3$, homogenization turns the realized sign vectors into the covectors of a realizable oriented matroid. Oriented matroids abstract the incidence, sidedness, and combinatorial duality of

hyperplane arrangements [FL78]. They are therefore a natural combinatorial shadow of an affine PolyTwist cut system.

They do not determine metric offsets, lengths, clearances, piece congruence, or swept collision. Two geometrically different realizations can share an oriented matroid. PolyTwist should preserve the oriented-matroid structure as an invariant, not mistake it for the mechanism.

3.5 Canonical strata versus computational cells

A connected sign component need not be a ball or a smooth manifold. Exact algorithms may refine the canonical stratification with a compatible semialgebraic triangulation or cylindrical algebraic decomposition [BCR98; ACM84]. Such a refinement is a certificate-bearing computational device. The canonical result is obtained by quotienting refinement cells back to their connected sign components and closure incidence.

Design rule

Never make a particular triangulation, mesh tessellation, or CAD projection order part of puzzle identity. Those are compiler choices. The sign-component structure is the invariant target.

Chapter 4

Traces, seams, pieces, and bonds

4.1 Raw traces and visible seams

The oriented carrier of cut i is

$$S_i = B \cap \phi_i^{-1}(0).$$

Its raw boundary trace is

$$T_i = S_i \cap \partial B.$$

The strata contained in T_i , together with closure incidence, encode trace components, intersections, endpoints, and topology.

A raw carrier trace is not automatically a visible seam. If the chambers on its two sides are bonded into one physical piece, the interface is mechanically absent even though the carrier was useful during construction.

Definition 4.1 (Visible mechanical seam [DEF-TRACE-001]). At a regular point of T_i with one adjacent three-dimensional chamber on each side, the point is a *visible mechanical seam* when those chambers belong to different physical pieces. The global seam is the closure of the set of such regular points. Decorative lines and stickers belong to $\mathcal{O}$, not to this seam set.

4.2 Exact affine trace clipping

Suppose a cut plane and a polyhedral boundary face lie in

$$n^T x = d, \quad m^T x = h,$$

with $v = n \times m \neq 0$. Define $D = \|v\|^2$ and

$$\alpha = \frac{d\|m\|^2 - h(n^T m)}{D}, \quad \beta = \frac{h\|n\|^2 - d(n^T m)}{D}, \quad x_0 = \alpha n + \beta m.$$

Then the plane intersection is $x(t) = x_0 + tv$. If the face has inequalities $c_r^T x \leq e_r$, each gives

$$(c_r^T v)t \leq e_r - c_r^T x_0.$$

Positive coefficients give upper bounds, negative coefficients give lower bounds, and zero coefficients give feasibility tests.

Theorem 4.2 (Exact trace clipping [THM-TRACE-CLIP-001]). *For an affine cut and a polyhedral boundary face, the trace is empty, a point, a segment, a ray, a line, or the entire face in a degenerate coincident case. In the bounded-face case with nonparallel planes, its segment endpoints are $x(L)$ and $x(U)$, where $[L, U]$ is the exact intersection of the induced scalar intervals. Its length is $(U - L)\|v\|$.*

Proof. The two independent plane equations define the affine line $x_0 + \mathbb{R}v$; direct substitution verifies $n^T x_0 = d$ and $m^T x_0 = h$. Restricting every face half-space to that line yields a one-dimensional linear inequality. Their intersection is an interval, possibly empty or degenerate. The Euclidean displacement between endpoints is $(U - L)v$. $\square$

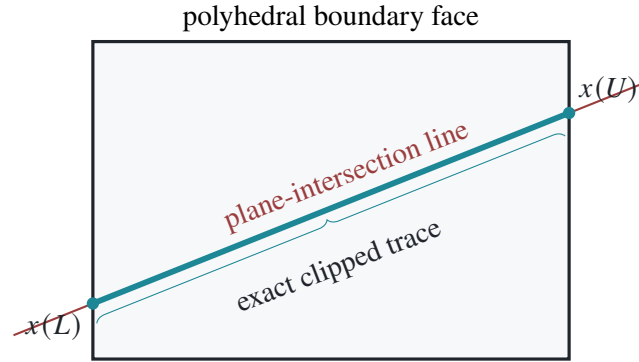


Figure 4.1: All trace geometry is line clipping in the affine-polyhedral specialization. Pixels are not invited to estimate the endpoints afterward.

The exact angle between a cut and face is represented without floating inverse trigonometry by the normalized pair

$$\left(n^T m, \|n \times m\| \right),$$

which determines cosine and sine once normal lengths and orientations are fixed.

4.3 Bond systems and physical pieces

Let C be the finite set of atomic chambers. A bond system may be represented by a graph $G_\beta = (C, E_\beta)$ whose edges join chambers that are rigidly fused across a shared codimension-one interface. More general hyperedges are reducible to a graph by adding a hidden bond node, so the graph model loses no finite equivalence information.

Definition 4.3 (Physical piece [DEF-PIECE-001]). For a connected component K of G_β , define

$$Q_K = \text{cl} \left(\bigcup_{C \in K} C \right).$$

The bond system is *admissible* when every Q_K is a connected regular-closed semialgebraic solid, the interiors of distinct Q_K are disjoint, and their union is B .

Admissibility is not an aesthetic preference. A bond file that joins two remote chambers through empty space does not magically manufacture a rigid part; it manufactures an invalid specification.

Choose an arbitrary orientation for every edge of G_β and let D_β be its edge-by-chamber incidence matrix. For a chamber support vector $m \in \{0, 1\}^C$, an edge row evaluates to the difference of the selector values at its endpoints.

Theorem 4.4 (Bond-kernel criterion [THM-BOND-001]). *A binary selector m is constant on every physical piece if and only if*

$$D_\beta m = 0.$$

Proof. If $D_\beta m = 0$, the endpoints of every bond edge receive equal values. Equality propagates along every path, so m is constant on each connected component. Conversely, a componentwise constant selector has equal endpoint values on every edge, hence every incidence row vanishes. $\square$

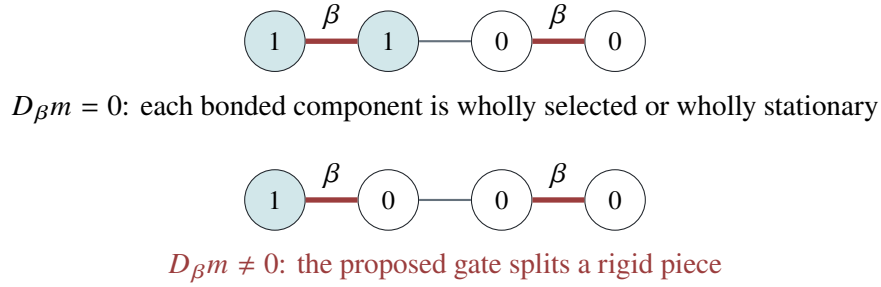


Figure 4.2: Bandage legality is a graph-incidence invariant, not an exception hard-coded beside the move parser.

4.4 Relation to the previous support condition

If a legal action has already been compiled into a permutation matrix P_a on finite slots, a selected support is a union of permutation cycles exactly when $P_a m = m$. That theorem remains correct. It is downstream of geometry. The bond-kernel criterion applies before an action permutation exists and therefore belongs in the foundation.

Chapter 5

Configuration space and exact observations

5.1 Piece placements and symmetries

Let $\mathcal{P}$ be the physical-piece set and let $Q_p \subset \mathbb{R}^3$ denote the home solid of piece p . A raw placement is

$$g = (g_p)_{p \in \mathcal{P}} \in \text{SE}(3)^{\mathcal{P}}.$$

If a piece has an intrinsic decorated symmetry group

$$H_p = \{h \in \text{SE}(3) : h(Q_p, \mathcal{O}_p) = (Q_p, \mathcal{O}_p)\},$$

then g_p and $g_p h$ represent the same placed decorated piece. A global rigid gauge may also be quotiented when no world-fixed frame is declared. Shape symmetry and decorated symmetry are deliberately distinct: a uniformly silver Mirror-Cube piece may have more observational symmetry than a labeled calibration piece.

5.2 Assembly constraints

The assembly data $\mathcal{J}$ contain exact semialgebraic predicates for anchors, joints, clearances, fixed bodies, and docking. Interior nonpenetration is not optional prose; it is part of the configuration predicate:

$$\text{int}(g_p Q_p) \cap \text{int}(g_q Q_q) = \emptyset \quad (p \neq q).$$

Fixed core bodies are treated as additional anchored solids and obey the same nonpenetration condition.

Definition 5.1 (Configuration and docked state spaces [DEF-CONF-001]). Let $\widetilde{\text{Conf}}(\mathfrak{P})$ be the set of raw placements satisfying all exact assembly predicates and interior nonpenetration. The *configuration space* is

$$\text{Conf}(\mathfrak{P}) = \widetilde{\text{Conf}}(\mathfrak{P}) / \left(G_{\text{gauge}} \times \prod_{p \in \mathcal{P}} H_p \right).$$

The *docked state space* $\Omega \subseteq \text{Conf}(\mathfrak{P})$ is the subset satisfying the docking predicate in $\mathcal{J}$.

The continuous space Conf is where a turn travels. The docked space Ω is where scrambles, plans, and benchmark states live. Confusing the two makes it impossible to state swept collision correctly.

5.3 The exact scene and its approximations

At state $x = [g] \in \Omega$, the exact visible scene is the union of transformed decorated boundaries

$$\mathcal{S}(x) = \bigcup_{p \in \mathcal{P}} g_p(\partial Q_p, O_p),$$

together with visible anchored bodies and any declared gaps or transparent components. Occlusion is determined by these transformed surfaces and the camera model.

A mesh $M_\varepsilon(x)$ is a certified approximation to $\mathcal{S}(x)$, ideally with a Hausdorff error bound, orientation consistency, and a topology certificate on strata where topology preservation is required. A rendered image is then

$$I = \text{Render}(M_\varepsilon(x), \text{camera}, \text{lighting}, \text{sensor}).$$

The image is an observation. It is not a second definition of x .

Design rule

Every public modality – RGB, depth, segmentation, normals, point clouds, tactile contacts, or action outcomes – must be generated from the same exact state and atlas. Cross-modal disagreement is a compiler defect, not “realistic noise,” unless an explicit noise model says otherwise.

5.4 Why the outer body is not a permanent container

The solved body B induces home piece shapes. A shape-modifying puzzle may leave the silhouette of B after a turn. Accordingly, admissibility is not the condition $\bigcup_p g_p Q_p = B$. It is the exact assembly relation in $\mathcal{J}$: nonpenetration, joints, anchors, clearances, and docking. This distinction permits Ghost and Axis mechanisms without granting every collection of flying shards diplomatic immunity from collision.

Chapter 6

Turn templates and exact legality

6.1 Gates and state-indexed anchors

A cut carrier explains where pieces came from. A turn gate explains which currently placed material is intended to move. The two often coincide in a solved state and need not coincide after other turns.

Definition 6.1 (Turn template [DEF-TURN-001]). A reversible turn template $a \in \mathcal{T}$ consists of:

1. a semialgebraic anchor-frame map $F_a : \Omega \rightarrow \text{SE}(3)$, frozen at the start of a turn;
2. a continuous semialgebraic gate function $\gamma_a : \mathbb{R}^3 \rightarrow \mathbb{R}$, whose positive side is selected;
3. a piecewise-semialgebraic rigid path $\rho_a : [0, 1] \rightarrow \text{SE}(3)$ with $\rho_a(0) = I$;
4. an exact endpoint docking predicate Δ_a ;
5. a reverse template $\bar{a}$ whose path is the time reversal of a and whose anchor/gate data are compatible at the endpoint.

For an atomic chamber C contained in physical piece $p(C)$ and a representative placement g of state x , its start-of-turn anchor coordinates are

$$C_x^a = F_a(x)^{-1} g_{p(C)} C.$$

Definition 6.2 (Gate atomicity and support). The gate is *atomic at x* when, for every atomic chamber C , the function γ_a has one strict sign throughout $\text{int } C_x^a$. The support vector $m_x(a) \in \{0, 1\}^C$ is

$$m_x(a)_C = \begin{cases} 1, & \gamma_a > 0 \text{ on } \text{int } C_x^a, \\ 0, & \gamma_a < 0 \text{ on } \text{int } C_x^a. \end{cases}$$

Gate atomicity rules out the splendid engineering strategy of rotating half a rigid chamber and hoping the mesh renderer does not notice.

6.2 The induced placement path

Assume gate atomicity and bond closure. A physical piece is selected when its chambers have support one. With $F = F_a(x)$, define

$$g_p(t) = \begin{cases} F \rho_a(t) F^{-1} g_p, & p \text{ selected,} \\ g_p, & p \text{ stationary.} \end{cases}$$

This gives a raw placement path $g(t)$. It descends to a path in configuration space precisely when all assembly constraints hold for every t .

Definition 6.3 (Exact legal turn [DEF-LEGALITY-001]). A turn template a is legal at $x \in \Omega$ when:

1. **gate atomicity** holds;
2. **bond closure** holds: $D_\beta m_x(a) = 0$;
3. **swept-path admissibility** holds: $[g(t)] \in \text{Conf}(\mathfrak{P})$ for all $t \in [0, 1]$;
4. **endpoint docking** holds: $[g(1)] \in \Omega$ and $\Delta_a(x, [g(1)])$ is true.

The domain $D_a \subseteq \Omega$ is the set of states where these conditions hold.

Theorem 6.4 (Legality factorization [THM-LEGALITY-FACTORIZATION-001]). *Under theorems 6.1 and 6.3, legal execution is equivalent to the conjunction of gate atomicity, bond closure, swept-path admissibility, and endpoint docking. Each failed factor admits an independent counterexample certificate.*

Proof. The equivalence is the definition. Independence follows from the certificate types: a straddling chamber witnesses gate failure; a bond edge with unequal selector values witnesses bond failure; a time and colliding point pair or violated joint equation witnesses path failure; and an endpoint equation or docking relation witnesses docking failure. None of these witnesses logically implies the others. $\square$

The useful object is therefore a structured residual

$$\Lambda_x(a) = (r_{\text{gate}}, D_\beta m_x(a), r_{\text{path}}, r_{\text{dock}}),$$

not one scalar whose zero somehow promises that geometry, topology, and mechanics have all behaved.

6.3 Semialgebraic path legality

Swept collision is a quantified statement. For two pieces p and q , collision during the proposed turn is expressed by

$$\exists t \in [0, 1] \exists u \in \text{int } Q_p \exists v \in \text{int } Q_q : g_p(t)u = g_q(t)v.$$

Joint and clearance violations are similar first-order formulas. With semialgebraic paths, all terms are semialgebraic.

Theorem 6.5 (Semialgebraic turn domains [THM-DOMAIN-SEMI-001]). *Assume B , all cuts, physical pieces, assembly relations, anchor maps, gates, motion paths, and docking predicates are semialgebraic. Then every turn domain $D_a \subseteq \Omega$ is semialgebraic.*

Proof. There are finitely many chamber support patterns. For a fixed pattern, gate atomicity is a finite collection of universally quantified sign conditions over semialgebraic chamber interiors. Bond closure is a finite linear equality over the binary pattern. Swept-path admissibility is a finite collection of universally quantified semialgebraic constraints in the time parameter and piece points. Docking is semialgebraic by assumption. Quantifier elimination over real closed fields converts each fixed-pattern formula to a quantifier-free semialgebraic formula; the domain is the finite union over admissible patterns. $\square$

This theorem gives completeness in principle, not cheap computation. General real quantifier elimination can exhibit doubly-exponential worst-case behavior [DH88]. The proper response is specialized exact geometry with a complete fallback, not a sudden spiritual attachment to epsilon tests.

6.4 Finite legality stratification

For a finite turn set $\mathcal{T} = \{a_1, \dots, a_r\}$, define the exact legal-action mask

$$\ell(x) = (\mathbf{1}_{D_{a_1}}(x), \dots, \mathbf{1}_{D_{a_r}}(x)) \in \{0, 1\}^r.$$

Theorem 6.6 (Finite legality stratification [THM-LEGALITY-STRATA-001]). *Under the hypotheses of theorem 6.5, Ω admits a finite partition into connected semialgebraic sets on each of which ℓ is constant.*

Proof. For each mask $b \in \{0, 1\}^r$, take the Boolean combination

$$L_b = \bigcap_{b_i=1} D_{a_i} \cap \bigcap_{b_i=0} (\Omega \setminus D_{a_i}).$$

It is semialgebraic. Each L_b has finitely many connected components, and there are finitely many masks. $\square$

The resulting components are the canonical *legality strata*. They are geometry-induced regions of state space, not bins invented to make a benchmark chart look industrious.

6.5 A selector example

For a half-space gate $\gamma(z) = n^T z - d$ and a convex chamber polytope C , gate atomicity is decided by the exact extrema

$$\min_{z \in C} \gamma(z), \quad \max_{z \in C} \gamma(z).$$

If the minimum is positive, C is selected; if the maximum is negative, it is stationary; otherwise the gate straddles the chamber. Once vertices are certified, these extrema occur at vertices. This is a precise affine fast path for the general quantified definition.

Chapter 7

Partial algebraic dynamics

7.1 Legal turns are partial bijections

For $x \in D_a$, let $\tau_a(x)$ be the docked endpoint of the legal path.

Theorem 7.1 (Turn partial bijection [THM-PARTIAL-BIJECTION-001]). *If $\bar{a}$ is the exact time-reversed template and the assembly constraints are time-symmetric, then*

$$\tau_a : D_a \longrightarrow D_{\bar{a}}$$

is a bijection with inverse $\tau_{\bar{a}}$.

Proof. Let $x \in D_a$ and write $y = \tau_a(x)$. Reversing the legal path preserves every time-symmetric assembly constraint and returns from y to x ; endpoint compatibility gives $y \in D_{\bar{a}}$ and $\tau_{\bar{a}}(y) = x$. The same argument with a and $\bar{a}$ exchanged proves both-sided inverse identities. $\square$

7.2 Inverse semigroup and legal-motion groupoid

Let $I(\Omega)$ be the symmetric inverse monoid of all partial bijections between subsets of Ω .

Theorem 7.2 (Inverse-semigroup dynamics [THM-INVERSE-SEMIGROUP-001]). *The partial bijections generated by $\{\tau_a : a \in \mathcal{T}\}$ and their inverses form an inverse subsemigroup*

$$S(\mathfrak{P}) \leq I(\Omega).$$

Proof. Partial bijections are closed under composition and inversion. Every element generated by the turn maps has a unique inverse as a partial bijection, so the generated subsemigroup is inverse. $\square$

This places state-dependent mechanisms in the established theory of partial actions and inverse semigroups [Exe98; KL04].

Definition 7.3 (Legal-motion groupoid). *The legal-motion groupoid is the action groupoid of germs associated with the inverse-semigroup action of $S(\mathfrak{P})$ on Ω . Its objects are docked states; an arrow records a locally defined legal transformation at a source state, and inversion is path reversal. For a finite discrete state space, the generator-labeled one-skeleton is exactly the legal state graph.*

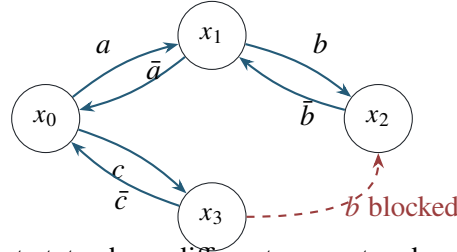
The inverse semigroup identifies move words that induce the same partial transformation. The execution path groupoid may retain the words themselves when benchmark provenance needs to distinguish two physically different routes with the same endpoint map.

7.3 Classical groups as a theorem, not an assumption

Theorem 7.4 (Group reduction [THM-GROUP-REDUCTION-001]). *If $D_a = \Omega$ for every primitive turn a , then every generator is a total permutation of Ω , $S(\mathfrak{P})$ is a group, and the generator-labeled state graph is a Schreier graph of that group action.*

Proof. A total partial bijection is an ordinary bijection. A semigroup generated by bijections and their inverses is a group. The labeled orbit graph of a group action with the chosen generators is its Schreier graph. $\square$

Classical unbandaged Rubik mechanics live here. Bandages, locks, geometry-dependent blocks, and jumbling generally do not.



Different states have different generator domains.
The graph is not regular, and forcing it into a total group action would encode illegal transitions.

Figure 7.1: A partial state graph. Dashed red denotes a proposed but undefined transition, not an edge to a magical “invalid” state.

7.4 Exact dynamical invariants

Natural invariants include:

- orbit decomposition of Ω ;
- idempotent domains in $S(\mathfrak{P})$;
- isotropy groups at states and strata;
- generator domains and their incidence;
- connected legality strata and their adjacency;
- labeled state-graph diameter, growth, bottlenecks, and automorphisms;
- experiment-outcome partitions induced by partial action sequences.

These are exact structures. A learned embedding can be useful later, but it does not become the mechanism because a t-SNE plot looked persuasive.

Chapter 8

Canonicity and the proof-carrying derivation architecture

8.1 Atlas isomorphism

Definition 8.1 (Exact atlas isomorphism). An isomorphism from $\mathfrak{P}$ to $\mathfrak{P}'$ consists of an orientation-preserving Euclidean isometry ψ , bijections of cut indices, chambers, pieces, and turn templates, and compatible maps of assembly and observation data such that:

1. $\psi(B) = B'$;
2. cut signs are preserved after cut reindexing:

$$\text{sgn } \phi'_{\pi(i)}(\psi x) = \text{sgn } \phi_i(x) \quad (x \in B);$$

3. bond incidence and physical-piece membership are preserved;
4. assembly predicates are conjugated by ψ and the piece relabeling;
5. each turn's anchor, gate, path, docking relation, and inverse are conjugated to the corresponding turn in $\mathfrak{P}'$;
6. declared observation data are preserved when the isomorphism is observation-sensitive.

Theorem 8.2 (Isomorphism invariance [THM-ISOMORPHISM-001]). *An exact atlas isomorphism induces:*

1. *an isomorphism of connected sign-component incidence posets;*
2. *congruences between corresponding physical pieces;*
3. *a homeomorphism of configuration and docked state spaces;*
4. *conjugacy of every primitive partial turn and of $S(\mathfrak{P})$;*
5. *an isomorphism of legal-motion groupoids and generator-labeled state graphs;*
6. *identical benchmark labels after the declared relabeling, and observation-equivalent renders under correspondingly transformed cameras.*

Proof. The sign-preservation condition maps each sign set to the corresponding sign set; a homeomorphism maps connected components and closure incidence bijectively. Bond preservation carries chamber components to chamber components, and ψ gives piece congruence. Conjugating placements by ψ preserves nonpenetration and all transported assembly predicates, giving the state-space homeomorphism. Turn conjugacy maps

each legal path and its four legality obligations to the corresponding legal path, hence $\tau'_{\pi(a)} = \Psi\tau_a\Psi^{-1}$ on the transported domain. The remaining statements follow functorially from conjugacy and deterministic rendering. $\square$

8.2 The exact derivation contract

The mathematical architecture has two synchronized branches:

$$\mathfrak{P} \mapsto \mathcal{X} \mapsto \mathcal{P} \mapsto (\text{Conf}, \Omega) \mapsto \{\tau_a\} \mapsto S(\mathfrak{P}), \mathcal{G}(\mathfrak{P}) \mapsto \text{state graph},$$

and

$$(\mathfrak{P}, x) \mapsto \text{exact transformed surfaces} \mapsto \text{certified mesh} \mapsto \text{observations}.$$

The branches meet at the same state x . Scramble generation, planning, and evaluation consume the state graph and exact state; image generation consumes exact transformed surfaces. Neither may reconstruct its own pieces from approximate meshes.

8.3 Proof-carrying compiler outputs

A future compiler should return an object and a certificate bundle. At minimum:

1. exact normalized coefficient data and sign-oracle provenance;
2. nonempty-stratum witnesses and sign certificates;
3. closure/facet incidence certificates;
4. piece-cover, disjoint-interior, and bond-component certificates;
5. gate extrema or quantified gate certificates;
6. bond-kernel vectors;
7. swept-collision proofs or explicit collision witnesses;
8. endpoint docking and reverse-turn certificates;
9. transition hashes derived from exact normalized state, never pixels;
10. mesh approximation certificates tied to exact strata.

An independent verifier should consume the bundle without trusting the compiler's internal search procedure.

8.4 Serialization is a separate problem

Plane equations can be normalized up to positive scale; algebraic numbers can use canonical minimal-polynomial and isolating-interval encodings; finite incidence structures can be canonically labeled. Global atlas symmetry makes the composition nontrivial.

Conjecture 8.3 (Practical canonical serialization [CONJ-CANONICAL-SERIALIZATION-001]). For bounded-degree algebraic atlases with finite symmetry groups and generic incidence, there exists a practical exact canonical-labeling pipeline whose serialized output is invariant under rigid atlas isomorphism and whose runtime is polynomial in the exact output size on generic instances.

No proof is offered. The mathematical isomorphism class is canonical now; a universally efficient byte string remains research.

8.5 Retired foundational abstractions

The earlier Slice-Action Code, restricted Walsh spectrum, and cut-action covariance may remain diagnostics on a finite compiled instance. They are not foundational because they compress away connected-component topology, physical piece geometry, path collision, and state-dependent domains. The Unified Legality Operator is replaced by the exact domain predicate and structured residual $\Lambda_x(a)$. The former notation survives only where it denotes a derived Boolean function, not the mechanism itself.

Chapter 9

Exact algorithms and complexity contracts

9.1 Algorithmic philosophy

The atlas theory is deliberately broader than the first compiler. A conforming implementation should therefore have a complete exact semantics, fast specialized paths, and an independently checkable fallback. “It rendered without exploding” is a user-interface test, not a correctness proof.

The principal algorithmic objects are finite whenever the requested docked component is finite, but their construction costs differ sharply:

- affine arrangements in fixed dimension admit efficient exact combinatorial algorithms;
- curved semialgebraic cuts are decidable through real algebraic geometry but can be expensive;
- axial polyhedral collision has exploitable structure that generic quantifier elimination ignores;
- exhaustive state graphs are necessarily output-sensitive because the graph itself may be enormous.

9.2 Affine-polyhedral compilation

Let the body be a bounded polyhedron with h supporting planes and let there be m affine cuts. Build the arrangement of all $N = m + h$ planes globally, rather than recomputing every possible cell by independent triple-plane enumeration. Classical arrangement algorithms construct an arrangement of N hyperplanes in fixed dimension d in $O(N^d)$ time in their model, which is optimal in the worst case because the output can have that size [EOS86]. In three dimensions the target bound is $O(N^3)$ plus exact coefficient costs and certificate output.

A deterministic affine compiler can proceed as follows:

1. normalize every plane equation over $\mathbb{K}$ while preserving orientation;
2. construct the global arrangement and its face-incidence structure;
3. retain cells lying in B and attach ternary cut signs;
4. merge arrangement cells that belong to the same connected sign component when body boundaries introduce a refinement;
5. derive atomic chamber adjacency from shared two-dimensional strata;
6. apply the bond quotient and validate physical pieces;
7. derive boundary traces by theorem 4.2;
8. compile turn gates and generate exact legality certificates.

For a boundary face described by k inequalities, one nonparallel cut trace is clipped in $O(k)$ exact comparisons once the line is known. All segment endpoints and lengths are algebraic whenever the input is algebraic.

9.3 General semialgebraic compilation

For polynomial or rational cut functions, a sign-invariant cylindrical algebraic decomposition or another certified semialgebraic cell decomposition refines all signs [Col75; ACM84; BPR06]. Connected-component reconstruction then produces the canonical stratification. A compatible triangulation supplies explicit simplices for rendering and topology checks without becoming canonical identity.

The complete route is intentionally uncompromising:

formula $\longrightarrow$ sign-invariant cells $\longrightarrow$ adjacency $\longrightarrow$ connected components $\longrightarrow$ certificates.

Approximate surface extraction may accelerate visualization, but it cannot replace the sign and adjacency oracle used by mechanics.

9.4 Gate classification

For a convex polyhedral chamber and affine gate, exact linear optimization decides the sign range. With certified vertices, scanning the vertices costs $O(v_C)$ for chamber C . Without vertices, fixed-dimensional linear programming is available.

For a general semialgebraic chamber C and gate γ , the three outcomes are formulas:

$$\begin{aligned} \text{selected} &\iff \forall z \in \text{int } C, \gamma(z) > 0, \\ \text{stationary} &\iff \forall z \in \text{int } C, \gamma(z) < 0, \\ \text{straddling} &\iff \exists z_+, z_- \in \text{int } C : \gamma(z_+) > 0 > \gamma(z_-). \end{aligned}$$

Quantifier elimination is complete. Specialized monotonicity, convexity, and interval methods should discharge ordinary cases first.

9.5 Swept collision

A complete verifier may encode collision as an existential formula over time and body coordinates. A useful implementation should exploit more structure:

- static separating planes preserved by axial motion;
- convex decomposition and support functions;
- exact continuous collision events between polyhedral features;
- symmetry and repeated piece geometry;
- interval subdivision with algebraic root isolation;
- cached certificates transported by atlas automorphisms.

Boundary of the claim

Open problem [OPEN-PATH-CERT-001]. Develop a sound, complete, output-sensitive exact collision algorithm specialized to axial polyhedral turns, with explicit degeneracy handling and a small

independently verifiable certificate. Generic CAD remains the completeness fallback, not the desired common path.

9.6 State-graph exploration

Given an exact transition oracle, breadth-first exploration of a reachable finite component is output-sensitive:

$$O(|V_R| + |E_R|)$$

transition-oracle calls and storage for the explored subgraph, excluding certificate sizes. Canonical state hashing must use exact normalized placements and declared symmetry quotients. A render digest is not a state hash; two occluded states can share pixels, a fact that cameras discovered long before benchmark designers did.

For shortest paths in an unweighted graph, BFS is exact. Weighted physical costs require Dijkstra or A* with an admissible lower bound. If legality is partial, illegal generators are absent edges rather than transitions to a fabricated sink state.

9.7 Complexity ledger

Task	Exact target	Complexity statement
Affine arrangement in $\mathbb{R}^3$	Global incidence complex	Classical $O(N^3)$ arrangement construction in fixed dimension, plus arithmetic and certificate cost.
One affine trace on one face	Segment or emptiness certificate	$O(k)$ clipping comparisons for a k -inequality face after line construction.
Bond quotient	Physical-piece components	$O(C + E_\beta)$ graph traversal; selector check $O(E_\beta)$.
Affine gate on convex chambers	Support or straddling witness	Linear in certified chamber vertices, or fixed-dimensional LP.
General semialgebraic strata	Connected sign components	Decidable through exact real algebraic geometry; no blanket practical polynomial bound is claimed.
General swept collision	Legal path or collision witness	Decidable by quantified semialgebraic formulas; generic worst case can inherit doubly-exponential phenomena.
Reachable finite state graph	Nodes, legal labeled edges	Output-sensitive BFS in explored vertices and edges times transition-oracle cost.
Canonical atlas serialization	Isomorphism-invariant bytes	Conjectural practical pipeline; symmetry-heavy worst cases unresolved.

Table 9.1: Complexity claims, with the optimism removed before it ferments.

9.8 Independent verification

The verifier should be smaller than the compiler and should check:

1. exact coefficient encodings and normalized signs;
2. every chamber witness and every claimed empty sign pattern;

3. incidence consistency and Euler/topology checks where applicable;
4. piece cover, disjoint interiors, and bond connectivity;
5. gate support and bond kernel;
6. collision certificates or counterexamples;
7. endpoint state, inverse transition, and exact state hash;
8. mesh-to-stratum correspondence and approximation bounds.

The research platform should be capable of rejecting its own compiler output. Otherwise “independent verification” is just the same bug wearing a second hat.

Chapter 10

Two exact rational ghost-like cut atlases

The examples in this chapter are exact cut bodies, not claims that cuts alone define complete commercial puzzles. Their purpose is to demonstrate rational mechanism frames, displaced carriers, connected chamber witnesses, exact trace geometry, and the separate addition of a legal turn template.

Throughout, world coordinates are $(X, Y, Z)^T$ and mechanism coordinates are

$$y = R^T(x - o).$$

The columns of R are the mechanism axes.

10.1 Rational Ghost Atlas A

Let

$$B_A = \left[-\frac{8}{5}, \frac{8}{5} \right]^3, \quad o_A = \begin{pmatrix} \frac{1}{12} \\ \frac{1}{15} \\ -\frac{1}{20} \end{pmatrix},$$

and

$$R_A = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

A direct multiplication gives $R_A^T R_A = I$ and $\det R_A = 1$. For each mechanism coordinate y_j , place cuts at $y_j = \pm \frac{1}{2}$.

10.1.1 Primitive world equations

10.1.2 Twenty-seven exact chamber witnesses

For each coordinate choose

$$y_j \in \left\{ -\frac{3}{4}, 0, \frac{3}{4} \right\},$$

according to whether the desired strict sign interval is below, between, or above the two cuts. Set $x = o_A + R_A y$. Every row of R_A has absolute row sum $5/3$, so

$$|X_k| \leq \max_i |o_{A,i}| + \frac{3}{4} \cdot \frac{5}{3} \leq \frac{1}{12} + \frac{5}{4} = \frac{4}{3} < \frac{8}{5}.$$

Thus all $3^3 = 27$ strict sign patterns have an explicit point inside B_A . By theorem 3.4, they are exactly the atomic chambers.

Axis	Level	Primitive world-coordinate plane
1	$-\frac{1}{2}$	$4X + 2Y - 4Z = -3$
1	$+\frac{1}{2}$	$4X + 2Y - 4Z = 3$
2	$-\frac{1}{2}$	$20X + 40Y + 40Z = -29$
2	$+\frac{1}{2}$	$20X + 40Y + 40Z = 31$
3	$-\frac{1}{2}$	$40X - 40Y + 20Z = -23$
3	$+\frac{1}{2}$	$40X - 40Y + 20Z = 37$

Table 10.1: Exact cuts for Rational Ghost Atlas A.

10.1.3 Exact traces on the face $X = 8/5$

Use (Y, Z) as coordinates on the face. Exact clipping gives the following segments.

Cut	Endpoint 1 (Y, Z)	Endpoint 2 (Y, Z)	Length
1–	$(-\frac{8}{5}, \frac{31}{20})$	$(-\frac{3}{2}, \frac{8}{5})$	$\frac{\sqrt{5}}{20}$
1+	$(-\frac{8}{5}, \frac{1}{20})$	$(\frac{3}{2}, \frac{8}{5})$	$\frac{31\sqrt{5}}{20}$
2–	$(-\frac{8}{5}, \frac{3}{40})$	$(\frac{3}{40}, -\frac{8}{5})$	$\frac{67\sqrt{2}}{40}$
2+	$(-\frac{8}{5}, \frac{63}{40})$	$(\frac{63}{40}, -\frac{8}{5})$	$\frac{127\sqrt{2}}{40}$
3–	$(\frac{11}{8}, -\frac{8}{5})$	$(\frac{8}{5}, -\frac{23}{20})$	$\frac{9\sqrt{5}}{40}$
3+	$(-\frac{1}{8}, -\frac{8}{5})$	$(\frac{59}{40}, \frac{8}{5})$	$\frac{8\sqrt{5}}{5}$

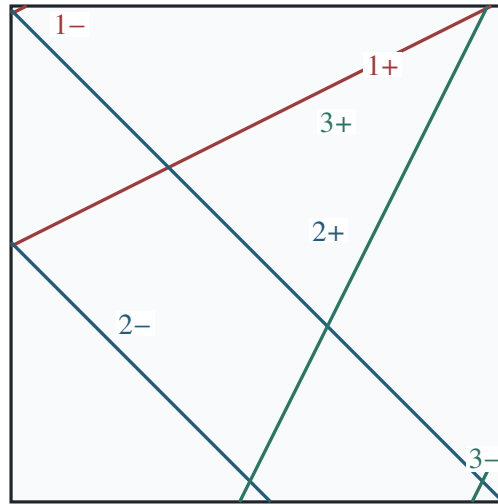
Table 10.2: Exact trace segments for Rational Ghost Atlas A on $X = 8/5$.

Figure 10.1: Solved-state cut traces of Rational Ghost Atlas A on the world face $X = 8/5$. The short 1– and 3– traces are not numerical accidents; their exact lengths are in table 10.2.

Proposition 10.1 (Exact specimen A [EX-GHOST-A-001]). *The data above define an affine-convex cut body with an exact rational orthogonal mechanism frame, exactly twenty-seven atomic chambers, and the six trace segments in table 10.2.*

Proof. Orthogonality and determinant are direct rational matrix calculations. The witness construction proves all twenty-seven strict sign regions nonempty; affine convexity proves connectedness and exhaustiveness. The trace table follows from theorem 4.2 by exact substitution into the square face inequalities. $\square$

10.2 Rational Ghost Atlas B

Let

$$B_B = \begin{bmatrix} -9 & 9 \\ -5 & 5 \end{bmatrix} \times \begin{bmatrix} -8 & 8 \\ -5 & 5 \end{bmatrix} \times \begin{bmatrix} -7 & 7 \\ -4 & 4 \end{bmatrix}, \quad o_B = \begin{pmatrix} -\frac{1}{18} \\ \frac{1}{14} \\ \frac{1}{21} \end{pmatrix},$$

and

$$R_B = \begin{pmatrix} \frac{1}{3} & -\frac{2}{15} & \frac{14}{15} \\ \frac{2}{3} & \frac{11}{15} & -\frac{2}{15} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{pmatrix}.$$

Again $R_B^T R_B = I$ and $\det R_B = 1$. Use the asymmetric cut levels

mechanism axis	lower cut	upper cut
1	$-\frac{2}{5}$	$\frac{3}{5}$
2	$-\frac{1}{3}$	$\frac{5}{3}$
3	$-\frac{3}{5}$	$\frac{1}{3}$

10.2.1 Primitive world equations

Axis	Level	Primitive world-coordinate plane
1	$-\frac{2}{5}$	$630X + 1260Y - 1260Z = -761$
1	$+\frac{3}{5}$	$630X + 1260Y - 1260Z = 1129$
2	$-\frac{1}{3}$	$-126X + 693Y + 630Z = -386$
2	$+\frac{5}{3}$	$-252X + 1386Y + 1260Z = 929$
3	$-\frac{3}{5}$	$882X - 126Y + 315Z = -610$
3	$+\frac{1}{3}$	$882X - 126Y + 315Z = 272$

Table 10.3: Exact cuts for Rational Ghost Atlas B.

10.2.2 Twenty-seven exact chamber witnesses

Choose each mechanism coordinate from $\{-\frac{4}{5}, 0, \frac{4}{5}\}$ according to its interval. The absolute row sums of R_B are $7/5$, $23/15$, and $5/3$. Therefore

$$\begin{aligned} |X| &\leq \frac{1}{18} + \frac{4}{5} \cdot \frac{7}{5} = \frac{529}{450} < \frac{9}{5}, \\ |Y| &\leq \frac{1}{14} + \frac{4}{5} \cdot \frac{23}{15} = \frac{1363}{1050} < \frac{8}{5}, \\ |Z| &\leq \frac{1}{21} + \frac{4}{5} \cdot \frac{5}{3} = \frac{29}{21} < \frac{7}{4}. \end{aligned}$$

All twenty-seven strict sign patterns are nonempty and hence are the atomic chambers.

10.2.3 Exact traces on the face $Y = -8/5$

Use (X, Z) as face coordinates.

Cut	Endpoint 1 (X, Z)	Endpoint 2 (X, Z)	Length
1−	$(-\frac{95}{63}, -\frac{7}{4})$	$(\frac{9}{5}, -\frac{121}{1260})$	$\frac{521\sqrt{5}}{315}$
1+	$(\frac{94}{63}, -\frac{7}{4})$	$(\frac{9}{5}, -\frac{2011}{1260})$	$\frac{97\sqrt{5}}{630}$
2−	$(-\frac{9}{5}, \frac{248}{315})$	$(\frac{9}{5}, \frac{2374}{1575})$	$\frac{18\sqrt{26}}{25}$
2+	No intersection with this face		0
3−	$(-\frac{27257}{17640}, \frac{7}{4})$	$(-\frac{5207}{17640}, -\frac{7}{4})$	$\frac{\sqrt{221}}{4}$
3+	$(-\frac{9617}{17640}, \frac{7}{4})$	$(\frac{12433}{17640}, -\frac{7}{4})$	$\frac{\sqrt{221}}{4}$

Table 10.4: Exact trace segments for Rational Ghost Atlas B on $Y = -8/5$.

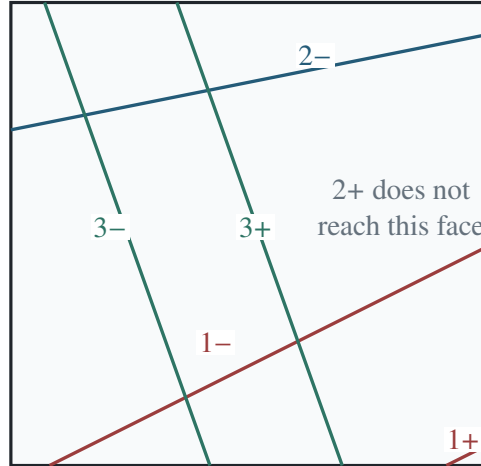


Figure 10.2: Solved-state cut traces of Rational Ghost Atlas B on $Y = -8/5$. A carrier can be globally present and locally invisible on a chosen face.

Proposition 10.2 (Exact specimen B [EX-GHOST-B-001]). *The data above define an affine-convex cut body with an exact rational orthogonal frame, exactly twenty-seven atomic chambers, the five listed face traces, and an exact empty trace for cut 2+ on $Y = -8/5$.*

Proof. The frame identities and chamber witness bounds are exact rational calculations. Exact line clipping yields the five segments. For cut 2+, the intersection line with $Y = -8/5$ violates the rectangular face bounds at both possible boundary crossings, so its clipping interval is empty. $\square$

10.3 Adding one exact turn without pretending the cuts chose it

For specimen A, let u_1 be the first column of R_A . Consider the gate

$$\gamma(x) = u_1^T(x - o_A) - \frac{1}{2}$$

and a quarter-turn path about the axis $o_A + \mathbb{R}u_1$, parameterized by theorem 2.2. Use no bonds and declare quarter-turn endpoints docked in an ideal pin-joint assembly.

Proposition 10.3 (Solved-state cap legality). *At the solved state of specimen A, the positive outer-cap turn satisfies gate atomicity, bond closure, and swept nonpenetration between selected and stationary pieces.*

Proof. Every atomic chamber lies strictly in one of the three u_1 coordinate intervals, so the gate boundary is a union of chamber faces and gate atomicity holds. There are no bonds. Rotation about u_1 preserves the scalar $u_1^T(x - o_A)$, hence selected interiors remain in the half-space $> 1/2$ and stationary interiors remain in the half-space $< 1/2$ throughout the path. Their interiors cannot meet. Selected pieces move by one common rigid motion, so their mutual disjointness is preserved; stationary pieces do not move. Endpoint docking is supplied separately by the declared ideal joint. $\square$

This proposition illustrates the division of labor. The cut identifies an exact gate that is legal for one declared joint. It does not prove that every apparent slice is a turn, nor that a commercial Ghost Cube has the same clearances. Those belong in $\mathcal{J}$ and $\mathcal{T}$.

Chapter 11

From mechanism to benchmark

11.1 The state graph is derived, not declared

For a finite docked component $V \subseteq \Omega$ and primitive turn set $\mathcal{T}$, define the labeled graph

$$G(\mathfrak{P}; V) = (V, E), \quad (x, a, y) \in E \iff x \in D_a \text{ and } y = \tau_a(x).$$

Every edge carries the four legality certificates from theorem 6.3. A benchmark may withhold those certificates from the model while retaining them privately for evaluation.

Planning is shortest-path or minimum-cost search on this graph or on a lazily generated exact subgraph. A plan is invalid if any proposed generator is outside its current domain, even when the token sequence would be valid on a visually similar ordinary cube.

11.2 Exact finite-memory scramble laws

A scramble process may depend on recent actions, controller state, or a no-immediate-inverse rule. Let C be a finite controller-memory set. For each product state (x, c) , let a rational policy assign probability to legal action and next-memory pairs.

Theorem 11.1 (Product-automaton scramble law [THM-PRODUCT-AUTOMATON-001]). *For a finite state graph and finite controller memory, the scramble process is a Markov chain on $V \times C$. Its length- L distribution is an exact product of transition matrices. If all policy probabilities are rational, every finite-horizon state probability is rational.*

Proof. The pair (x, c) contains all information needed to sample the next legal action and controller state, so it is Markov. Let P_t be the product-state transition matrix at step t . From initial row distribution μ_0 , the distribution after L steps is

$$\mu_L = \mu_0 P_1 P_2 \cdots P_L,$$

or $\mu_0 P^L$ in the homogeneous case. Matrix addition and multiplication preserve rationality. □

This retains the strongest part of the previous ledger’s transfer-matrix formalism while grounding its states and transitions in the atlas.

11.3 Passive observations cannot identify every mechanism

Theorem 11.2 (Boundary nonidentifiability [THM-BOUNDARY-NONID-001]). *There exist two exact atlases with identical solved-state exterior body, boundary cut traces, and exterior appearance, but different internal chamber and mechanism structures. Therefore a passive observation of the exterior at one state cannot identify the atlas in general.*

Proof. Let $B = [-1, 1]^3$. Atlas A_0 has no cut carriers and one physical piece. Atlas A_1 adds the internal spherical carrier

$$X^2 + Y^2 + Z^2 = \frac{1}{4},$$

with the inner ball and outer shell unbonded. On ∂B , at least one coordinate has absolute value one, so $X^2 + Y^2 + Z^2 \geq 1$; the internal sphere has empty intersection with ∂B . Thus both atlases have the same exterior body and no boundary trace. Give their exterior surfaces the same appearance. Their chamber and piece counts differ, and A_1 may additionally assign a hidden internal joint to the inner piece. No one-state exterior observation distinguishes them. $\square$

The theorem is an impossibility result, not an invitation to surrender. It says that active action-outcome queries, multiple states, tactile channels, or priors are mathematically necessary for universal identification.

11.4 Finite exact system identification

Let $\mathcal{H}$ be a finite hypothesis set of atlases. An exact experiment e may be an admissible action sequence, adaptive policy, camera intervention, or multimodal query. In the noiseless setting it induces an outcome map

$$O_e : \mathcal{H} \rightarrow \mathcal{Y}_e.$$

Theorem 11.3 (Finite identifiability [THM-FINITE-ID-001]). *A finite deterministic hypothesis class is exactly identifiable by a finite adaptive experiment tree if and only if every distinct pair $h, h' \in \mathcal{H}$ is separated by some admissible experiment e , meaning $O_e(h) \neq O_e(h')$. The optimal worst-case number of experiments equals the minimum depth of a separating decision tree with singleton leaves.*

Proof. Necessity is immediate: hypotheses with identical outcomes under every experiment reach the same leaf in every decision tree.

For sufficiency, take any current candidate set S with $|S| > 1$ and choose two distinct hypotheses $h, h' \in S$. By assumption an experiment separates them, so its outcome partition splits S into proper subsets. Recurse on every nonempty subset. Cardinality decreases on every branch, so the finite recursion ends at singleton leaves. The final statement is the definition of optimal worst-case decision-tree depth. $\square$

For a prior $p(h)$, the expected entropy reduction of an experiment is

$$\text{IG}(e) = H(H) - \mathbb{E}_{Y_e} [H(H \mid Y_e)],$$

a classical Bayesian experimental-design quantity going back to Lindley [Lin56]. PolyTwist may contribute exact outcome partitions and mechanism hypotheses; it does not get to rename expected information gain and claim discovery.

11.5 Benchmark task spine

A next-generation benchmark should organize tasks by which derived map is hidden, not by how many labels can fit in a catalog.

Task family	Hidden object	Exact target
Geometric reconstruction	B , cut carriers, traces, incidence	Atlas-equivalence class or certified partial reconstruction; exact metric and topological errors.
Piece induction	Bond quotient and physical solids	Chamber-to-piece partition, piece geometry, adjacency, seam set.
Legality prediction	Turn domains D_a	Legal mask plus gate, bond, path, and docking diagnosis.
State tracking	Current placement class x	Exact state or orbit representative under declared observational symmetries.
Active identification	Hypothesis atlas h	Experiment policy, posterior, separating depth, calibrated uncertainty.
Planning	Labeled partial state graph	Legal plan, cost, optimality gap, certificate coverage.
Counterfactual reasoning	Modified atlas component	Correctly recompiled downstream consequences rather than local feature edits.

Table 11.1: Task families derived from the atlas.

11.6 Evaluation without arbitrary feature vectors

Primary metrics should be tied to exact objects:

- sign-component and incidence precision/recall under optimal atlas alignment;
- exact or tolerance-certified trace endpoint and length error;
- chamber-to-piece partition distance and seam-set error;
- legality accuracy decomposed by the four proof obligations;
- action-query complexity and decision-tree regret for identification;
- plan validity, cost, and optimality gap;
- calibration of posterior mass over atlas equivalence classes;
- invariance under camera, appearance, and exact atlas isomorphisms.

Secondary learned representations are welcome. They are evaluated against these structures rather than promoted into definitions because a benchmark needed a single leaderboard number.

11.7 Public/private separation

A benchmark episode should be generated from a private atlas and state. The public projection may reveal images, action aliases, partial traces, or query access. The private target retains exact atlas data, certificates, state, legal domains, and reference plans. Leakage tests should verify that canonical IDs, source cut coefficients, hidden action mappings, and exact state hashes do not enter public artifacts.

The evaluator recompiles or verifies targets from the private atlas. It should not trust a separately stored answer file when the answer is derivable from the canonical object.

Chapter 12

Novelty boundary and prior art

12.1 What is already established

The mathematical ingredients are not blank territory:

- hyperplane arrangements and their face complexes are classical [Zas75; EOS86];
- oriented matroids abstract arrangement signs and incidence [FL78];
- semialgebraic decomposition and quantifier elimination are established [Col75; BPR06];
- configuration-space collision and exact motion planning are established [Loz83; SS83a; SS83b];
- partial actions and inverse semigroups are established [Exe98; KL04];
- Rubik-type groups and algorithms are an extensive field, with asymptotic results far beyond benchmark folklore [Dem+11].

Software prior art is equally relevant. Twizzle/cubing.js PuzzleGeometry derives permutation, SVG, and 3D representations from symmetric flat cuts of Platonic solids, and its API exposes permutation and rendering outputs [cub26b; RGc26]. Its first-party help explicitly states that bandaging and jumbling are unsupported in that model [RGc26]. KPuzzle is a permutation-and-orientation execution representation [cub26a]. Magic Puzzle Ultimate already lets users specify symmetry groups, axes, cutting planes, twists, and body boundaries, including higher-dimensional mechanisms [Ast26].

12.2 Defensible contribution candidate

The plausible contribution is the exact synthesis:

1. a canonical connected sign-component representation of arbitrary semialgebraic cuts;
2. a bond quotient that derives physical pieces and visible seams;
3. an exact configuration/docking space for the assembled mechanism;
4. state-indexed gate selection and swept-path legality;
5. inverse-semigroup and groupoid dynamics derived from reversible legal turns;
6. a proof-carrying compilation chain into geometry, rendering, state graphs, scrambles, planning, and active identification;
7. representation-invariant benchmark evaluation.

This is a research direction, not a certified novelty verdict. A complete paper must continue searching patents, puzzle-design literature, mechanical CAD systems, robotics languages, general simulators, and

Prior line	Established capability	Remaining PolyTwist target
Arrangement / oriented-matroid theory	Cut incidence and combinatorial sidedness	Body-truncated connected sign components, physical bonds, metric geometry, and motion legality in one object.
PuzzleGeometry / Twizzle	Symmetric plane-cut puzzle generation, permutations, SVG/3D output	Arbitrary semialgebraic cuts, state-dependent path legality, bonds, jumbling, proof certificates, benchmark semantics.
KPuzzle / ksolve formats	Efficient finite permutation and orientation execution	Geometric derivation of pieces, domains, and legal transitions.
Magic Puzzle Ultimate	Broad user-defined axes, planes, twists, boundaries, higher dimensions	A formal exact semantics with theorems, independent verification, partial dynamics, and AI-evaluation derivation.
Robot motion planning	Configuration-space obstacles and exact planning	Specialized cut-and-turn structure, docking, chamber gates, and puzzle action groupoids.
Partial action theory	Algebra of partially defined transformations	Geometric derivation of those partial transformations from a physical atlas.

Table 12.1: The novelty boundary is an integration problem, not a contest to rename classical mathematics.

unpublished community tools. The source registry records what has actually been checked.

12.3 What this monograph does claim

It claims the repository-specific definitions and theorems listed in `research/claims.json`, including the cut-underdetermination theorem, bond-kernel criterion, legality factorization, semialgebraic turn-domain theorem, legality stratification, partial-bijection dynamics, group reduction, isomorphism invariance, passive nonidentifiability, product-automaton scramble law, and finite identifiability theorem.

It does not claim to have invented hyperplane arrangements, configuration spaces, expected information gain, inverse semigroups, plane-cut puzzle generation, or the general idea that a cube can be made unnecessarily difficult.

Chapter 13

Research roadmap and open problems

13.1 Mathematics-first promotion gates

Implementation should remain frozen until the mathematical schema passes the following gates.

Gate M1: exact atlas schema

- Formalize coefficient encodings, regular-closed bodies, cut functions, bonds, assembly constraints, turn templates, and observation data.
- Specify exact isomorphism and noncanonical display metadata separately.
- Provide at least cube, tetrahedral, dodecahedral, ghost-like, mirror-like, axis-like, bandaged, and jumbled examples on paper.

Gate M2: affine-polyhedral theorem package

- Global arrangement construction and exact trace clipping.
- Piece and seam derivation with independent certificates.
- Exact affine gates, bond closure, and quarter-turn path checks.
- Proof that legacy cubic instances embed faithfully into the atlas.

Gate M3: partial dynamics

- Exact docking and reverse-template semantics.
- Inverse-semigroup/groupoid export.
- State hashing modulo declared piece and global symmetries.
- Legal-mask strata and group-action reduction tests.

Gate M4: complete path legality

- Specialized polyhedral axial collision algorithm.
- Quantifier-elimination fallback and certificate verifier.
- Clearance, kerf, and tolerance semantics.
- Jumble examples where another axis becomes legal only at an intermediate docking state.

Gate M5: observation and benchmark functor

- Certified mesh derivation from exact strata.
- Public/private projections with leakage tests.
- Active-identification hypotheses generated by exact atlas transformations.
- Evaluation invariant under atlas isomorphism and observation nuisance factors.

13.2 Open mathematical and algorithmic problems

Boundary of the claim

Observation identifiability [OPEN-RECON-001]. Characterize atlas equivalence classes identifiable from one image, passive multi-view images, legal-action queries, action videos, depth, tactile contacts, and combinations thereof. Prove impossibility and positive reconstruction results under explicit occlusion and appearance models.

13.2.1 Curved and nonconvex cuts

Develop practical connected-component and incidence algorithms for bounded-degree algebraic carriers, especially when one sign condition has multiple components or nontrivial topology. CAD is complete but may be excessive; arrangement-like incremental methods for tame surfaces could exploit the puzzle setting.

13.2.2 Tolerance and robust legality

Physical puzzles have kerf, bevels, compliance, backlash, and manufacturing tolerance. A robust atlas should replace exact contact by quantified uncertainty sets and define:

surely legal, possibly legal, surely illegal.

The resulting domains may be described by erosion/dilation or robust optimization, but no theorem is claimed yet.

13.2.3 Canonical serialization

Resolve theorem 8.3. Important cases include repeated congruent pieces, highly symmetric bodies, equivalent polynomial cut descriptions, and mechanisms whose observation decoration breaks only part of the mechanical symmetry.

13.2.4 Groupoid invariants and difficulty

Determine which exact invariants predict planning complexity, active-identification depth, or multimodal reasoning difficulty. Candidate structures include domain semilattices, isotropy, legality-stratum adjacency, orbit growth, and bottleneck profiles. Any empirical scalar must be motivated by a theorem or a controlled scientific question, not harvested by feature-vector trawling.

13.2.5 Mechanism reconstruction

Given images or meshes of a manufactured puzzle, infer a posterior over atlases rather than a single guessed permutation model. The boundary nonidentifiability theorem shows why hidden cuts and joints require priors or interventions.

13.3 Paper-level falsification criteria

The framework should be considered refuted or substantially incomplete if any of the following occurs:

- a common cut-and-turn mechanism cannot be represented without an independent hand-coded transition table;
- two atlas-isomorphic inputs yield nonconjugate compiled dynamics;
- a legal move can pass the verifier while splitting a physical piece or colliding during the path;
- rendering requires a piece partition not derivable from the atlas;
- the proposed novelty collapses to an existing formalism after a complete literature comparison;
- benchmark performance can be solved by exploiting serialization or renderer artifacts rather than recovering mechanism structure.

A research framework should publish the conditions under which it is wrong. Otherwise it is branding with equations.

Chapter 14

Conclusion

The clean mathematical language for a cut-and-turn mechanism is not a single permutation group and not a bag of visual features. It is an exact stratified body with physical bonds, an assembled configuration space, and reversible partial motions. From that object, one obtains traces, pieces, legal domains, partial dynamics, state graphs, scrambles, plans, and observations.

The central structural move is simple:

$\text{cuts} \longrightarrow \text{connected strata} \longrightarrow \text{pieces} \longrightarrow \text{configurations} \longrightarrow \text{partial legal motions}$
--

The rest is disciplined derivation. Hyperplane arrangements explain the affine combinatorics. Semialgebraic geometry guarantees finiteness and decidability. Configuration space makes swept legality precise. Inverse semigroups and groupoids express state-dependent moves without forcing them into a total group. Exact experiment partitions make active identification scientific rather than theatrical.

PolyTwist can therefore become more than a simulator and more than a benchmark. It can become a proof-carrying language in which unfamiliar mechanisms are specified once and every downstream artifact is accountable to that specification. That is a considerably harder project than generating more cubes, which is usually how one notices it might be worth doing.

Appendix A

Notation and object map

Symbol	Meaning
$\mathbb{K}$	Effective exact ordered coefficient domain, normally real algebraic numbers.
B	Solved-state regular-closed body.
$\Phi = (\phi_i)$	Ordered oriented semialgebraic cut functions.
S_i	Cut carrier $B \cap \phi_i^{-1}(0)$.
T_i	Raw boundary trace $S_i \cap \partial B$.
$\mathcal{X}(B, \Phi)$	Connected sign-component stratification with closure incidence.
C	Atomic three-dimensional chambers.
β, G_β, D_β	Bond system, bond graph, and oriented incidence matrix.
$\mathcal{P}, Q_p$	Physical-piece set and home solid of piece p .
$\mathcal{J}$	Exact assembly, anchor, joint, clearance, symmetry, and docking data.
$\text{Conf}(\mathfrak{P})$	Continuous admissible configuration space.
Ω	Docked state space.
$\mathcal{T}$	Finite family of reversible turn templates.
$F_a, \gamma_a, \rho_a, \Delta_a$	Anchor, gate, rigid path, and docking predicate for turn a .
$m_x(a)$	Binary chamber support selected by turn a at state x .
D_a	Exact legality domain of turn a .
τ_a	Partial bijection induced by a legal turn.
$S(\mathfrak{P})$	Inverse semigroup generated by legal partial turns.
$\mathcal{G}(\mathfrak{P})$	Legal-motion action groupoid.
O	Appearance and sensor data.

Table A.1: Notation used in the monograph.

Appendix B

Claim and theorem index

Claim ID	Status	Location / content
THM-SIGN-FINITE-001	Proved	Theorem 3.2: finite canonical stratification.
THM-AFFINE-CONVEX-001	Proved	Theorem 3.4: strict signs suffice in the affine-convex case.
THM-TRACE-CLIP-001	Proved	Theorem 4.2: exact trace endpoints and lengths.
THM-CUTS-UNDERDET-001	Proved	Theorem 1.1: cuts alone do not define moves.
THM-BOND-001	Proved	Theorem 4.4: exact bond-closure criterion.
THM-LEGALITY-FACTORIZATION-001	Proved	Theorem 6.4: four legality obligations.
THM-DOMAIN-SEMI-001	Proved	Theorem 6.5: turn domains are semialgebraic.
THM-LEGALITY-STRATA-001	Proved	Theorem 6.6: finite constant-mask strata.
THM-PARTIAL-BIJECTION-001	Proved	Theorem 7.1: reversible turns are partial bijections.
THM-INVERSE-SEMIGROUP-001	Proved	Theorem 7.2: generated inverse semigroup.
THM-GROUP-REDUCTION-001	Proved	Theorem 7.4: total case becomes group action.
THM-ISOMORPHISM-001	Proved	Theorem 8.2: representation invariance.
THM-BOUNDARY-NONID-001	Proved	Theorem 11.2: passive exterior nonidentifiability.
THM-PRODUCT-AUTOMATON-001	Proved	Theorem 11.1: exact scramble matrices.
THM-FINITE-ID-001	Proved	Theorem 11.3: finite experiment identifiability.
EX-GHOST-A-001	Proved example	Section 10.1.
EX-GHOST-B-001	Proved example	Section 10.2.
CONJ-CANONICAL-SERIALIZATION-001	Conjecture	Theorem 8.3.
OPEN-PATH-CERT-001	Open	Specialized exact axial collision certificates.
OPEN-RECON-001	Open	Passive and active atlas reconstruction.
OPEN-CURVED-CUTS-001	Open	Certified compilation for curved and nonconvex cut carriers.
OPEN-TOLERANCE-001	Open	Robust semantics for clearance, contact, and manufacturing tolerance.
OPEN-GROUPOID-INVARIANTS-001	Open	Exact groupoid invariants for planning and identification difficulty.

Table B.1: Repository-specific claim map. The machine-readable source of truth is `research/claims.json`.

Appendix C

Audit checklist for a future implementation

A future atlas implementation should not be called complete until it can answer each item with a certificate or a precise unsupported-status marker.

1. Is every canonical coefficient exact, normalized, and sign-decidable?
2. Are cut carriers and orientations explicit?
3. Are connected sign components distinguished when one sign vector is disconnected?
4. Are trace components and all closure incidences exact?
5. Are physical pieces derived from chambers and bonds?
6. Are visible seams derived after the bond quotient?
7. Is the continuous configuration space distinct from the docked state space?
8. Does every turn provide an anchor, gate, exact path, docking relation, and reverse?
9. Can legality failures identify gate, bond, path, or docking causes separately?
10. Is swept collision checked for the whole path?
11. Are legal transitions partial bijections with verified inverses?
12. Does total legality reduce to the expected classical group action?
13. Are state hashes invariant under declared symmetries and independent of rendering?
14. Are meshes and pixels traceable to exact surfaces and states?
15. Are scrambles distributions on legal paths rather than unverified token strings?
16. Are benchmark labels re-derived or independently verified from the atlas?
17. Are external sources linked to specific claims in the source registry?
18. Are conjectures, evidence, and proved claims visibly separated?

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